

COURSE DESCRIPTION			
Program	Fire Technology and Safety		
Course Title	Computer Applications in Fire Technology & Safety		
Course Code	2469	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

This course helps to improve the software knowledge of students for applying in designing and drafting of 2D components.

Course Objectives

1	To familiarize the different software used for drawing
2	To Introduce the basic commands and tools necessary for two-dimensional drawing, design and drafting using AutoCAD
3	To apply knowledge of AutoCAD for drawing simple fire safety drawing components
4	To introduce CAD tools used in fire and safety system design

Course Pre-requisite

Topic	Course code	Course Title	Semester
Knowledge of Basic Engineering Drawing and Projections	1003	Engineering Graphics	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Illustrate the use of Computer Aided Drafting software and identify various commands used in CAD	Understand
CO2	Apply knowledge to draw simple two dimensional drawings and sections using CAD	Apply
CO3	Apply knowledge to draw simple circuit diagrams of fire safety components	Apply
CO4	Construct Isometric drawings of simple objects	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	2	-	-	-	-	-	-	-
CO2	3	-	-	3	-	-	-	-	-	-	-
CO3	3	-	-	3	-	-	-	-	-	-	2
CO4	3	-	-	3	-	-	-	-	-	-	-
Total	12	-	-	11	-	-	-	-	-	-	2
Total	3	-	-	2.75	-	-	-	-	-	-	2

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total
0	0	3	0	1.5	45	1.5				60	40	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Introduction to Computer Aided Drafting	Study about the Computer Aided Drafting software and its uses and applications	CO1	Understand	3
2	Basic commands and tools used in CAD	Study the basic commands and tools used in CAD software for drawing	CO1	Understand	3
Module II					
3	Drawing basic 2D entities using CAD	Draw basic 2D entities like rectangle, rhombus, circle, arc, polygon etc. using AutoCAD software	CO2	Apply	3
4	Drawing simple geometric figures using AutoCAD	Drawing blocks of 2D entities comprises of rectangle, circle, polygon, rectangular array, blocks using AutoCAD	CO2	Apply	3
5	Draw orthographic views of simple engineering objects	Draw orthographic views of simple engineering objects like stepped block, c block etc. using AutoCAD(Any 1)	CO2	Apply	3
6	Draw sectional views of simple objects	Draw the sectional views of simple objects using various tools used in AutoCAD	CO2	Apply	3
	Series Exam - I		CO2	Apply	3
Module III					
7	Draw the simple circuit diagram of typical control unit of fire sprinkler system	Apply the knowledge of various tools used in AutoCAD for drawing simple sketches of fire safety components	CO3	Apply	3
8	Draw the simple circuit diagram of local fire alarm system	Apply the knowledge of various tools used in AutoCAD for drawing simple sketches of fire safety components	CO3	Apply	3
Module IV					
9	Draw the isometric views of simple objects	Apply knowledge about various tools used in AutoCAD for drawing the isometric view of simple objects like stepped block	CO4	Apply	6
10	Draw isometric views of simple fire safety components	Apply knowledge about various tools used in AutoCAD for drawing the isometric view of simple fire safety components	CO4	Apply	6
	Series Exam - II		CO4	Remember	3
	Open Ended Experiment		CO4	Remember	3

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviors.
Integrate multiple concepts from the syllabus into a combined application.
Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.
Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Recollect knowledge about engineering drawing. Understand the applications and uses of CAD software	1.5
2	Observe the various tools used in AutoCAD Practice the tools used in AutoCAD	1.5
3	Prepare lab record Refer youtube videos	1.5

4	Refer the commands used in CAD Practice the commands used in CAD	1.5
Module II		
5	Recollect the knowledge about basic drawing Prepare lab record	1.5
6	Understand the application of CAD tools Practice various CAD tools for developing simple figures	1.5
7	Prepare for lab exam Prepare lab record	1.5
8	Understand the various projections and sections Refer video lectures	1.5
Module III		
9	Refer various sample circuit diagrams of fire safety systems Refer videos on drawing	1.5
10	Prepare lab record Practice tools for drawing simple circuit diagrams	1.5
11	Refer video lectures Understand the fire safety components	1.5
Module IV		
12	Prepare lab record Refer various sample drawings in internet	1.5
13	Recollect knowledge of tools Understand isometric projection	1.5
14	Recollect knowledge of isometric projection in engineering drawing Prepare for lab exam	1.5
15	Refer for open ended experiment Open ended experiment	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Engineering Graphics	P I Varghese, K C John	VIP Publishers
2	Engineering Graphics with AutoCAD	Kulkarni D. M.; Rastogi A. P.; Sarkar A. K	PHI Learning Private Limited- New Delhi (2010); ISBN: 978

REFERENCE			
Sl. No.	Title	Author	Publication
1	Engineering Drawing	N. D Bhatt	Charotar Publishing House, Anand, Gujarat 2010; ISBN: 978-93- 80358-17-8.
2	Engineering Graphics with AutoCAD	Kulkarni D. M.; Rastogi A. P.; Sarkar A. K	PHI Learning Private Limited- New Delhi (2010); ISBN: 978
3	Essentials of Engineering Drawing and Graphics using AutoCAD	Jeyapoovan T	Vikas Publishing House Pvt. Ltd, Noida, 2011; ISBN: 978-8125953005
4	AutoCAD User Guide	Autodesk	Autodesk Press, USA, 2015

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	http://www.swayam.gov.in/	All
2	https://nptel.ac.in/course.html	I, II & IV
3	https://ndl.iitkgp.ac.in/	I, II & IV

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Computer	Intel Core i5 / AMD Ryzen 5 processor, 8 GB RAM (16 GB recommended), and a 256 GB SSD. A dedicated graphics card with minimum 2 GB VRAM and a Full HD (1920×1080) display is recommended for smooth drafting performance. For better performance in 3D modeling and larger drawings, use 16 GB RAM, 512 GB SSD, and a mid-range NVIDIA or AMD GPU with 4 GB VRAM or higher.	30
2	LCD Projector	Display Technology: LCD or DLP; Resolution: 1920 × 1080 (Full HD) minimum (better clarity for CAD details); Brightness: 4000–5000 lumens (for visibility even with ambient light); Contrast Ratio: ≥ 10,000:1 (better depth and detail); Aspect Ratio: 16:9 / 16:10; Throw Type: Standard or Short Throw (depends on room size); Input Ports: HDMI, VGA, USB (HDMI preferred); Lamp Life: 10,000+ hours (eco mode); Keystone Correction: Manual/Auto (helps quick setup); Speakers: Built-in (optional; external recommended for audio)	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	AutoCAD software	Full AutoCAD version (not AutoCAD LT) with both 2D and basic 3D modeling capabilities, running on 64-bit Windows with support for DWG/DXF formats and lab network licensing if needed.	30

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to		30	15	15	10	
Marks	5	30	15		10	40
Total marks		60				40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8

Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50

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